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☐ Vendor Study Protocol				☐ Vendor Study Report		☒ Lpath Study Report	
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Study Title: Efficacy of murine anti-LPA Ab in the STZ model of type 1 diabetes - Repeat Dose Study							
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Female Sprague-Dawley rats were rendered diabetic by a single IP injection of STZ and once allodynia and diabetes were confirmed, animals were randomized into treatment groups and s.c. treatments were initiated on week 5 (Day 0) of the study. Two dose levels of B3 (10 and 30 mg/kg) treated rats were compared to their corresponding IgG control, vehicle treated and non-diabetic animals. Animals were tested weekly for the presence of hyperalgesia and tactile allodynia 3 days after last dosed. On day 28 of the study all treatments were stopped and allodynia continued to be measured weekly until the end of the study. Results showed that anti-LPA antibody at 10 and 30 mg/kg increased the paw withdrawal threshold in diabetic rats when compared to IgG control or vehicle treated animals to a level similar to that observed in non-diabetic rats. This marked reduction in mechanical allodynia is observed when tested 3 days after drug delivery, indicating a sustained and specific effect of drug that washes out in approximately two weeks after treatments stopped. In, summary, anti-LPA antibody (B3) therapy alleviated established allodynia in a sustained manner in the rat STZ-induced model of diabetic neuropathy.							

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Efficacy of murine anti-LPA Ab in the STZ rat model of type 1 diabetes: Repeat Dose Study

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1 SUMMARY

This efficacy study in STZ induced diabetic rats was conducted at University of California San Diego (UCSD San Diego, CA). The purpose of this study was to determine the optimal dose level for the murine anti-lysophosphatidic acid (LPA) antibody (B3) to reach long term reversal of allodynia in the STZ rat model of type 1 diabetes.

Female Sprague-Dawley rats were rendered diabetic by a single IP injection of STZ and once allodynia and diabetes were confirmed, animals were randomized into treatment groups and s.c treatments were initiated on week 5 (Day 0) of the study. Two dose levels of B₃ (10 and 30 mg/kg) treated rats were compared to their corresponding IgG control, vehicle treated and non-diabetic animals. Animals were tested weekly for the presence of hyperalgesia and tactile allodynia 3 days after last dosed. On day 28 of the study all treatments were stopped and allodynia continued to be measured weekly until the end of the study. Results showed that anti-LPA antibody at 10 and 30 mg/kg increased the paw withdrawal threshold in diabetic rats when compared to IgG control or vehicle treated animals to a level similar to that observed in non-diabetic rats. This marked reduction in mechanical allodynia is observed when tested 3 days after drug delivery, indicating a sustained and specific effect of drug that washes out in approximately two weeks after treatments are stopped.

In summary, B3 therapy alleviates established allodynia in a sustained manner in the rat STZ-induced model of diabetic neuropathy.

2 INTRODUCTION

The bioactive lipid lysophosphatidic acid (LPA) is a key extracellular signaling molecule in neuropathic pain. LPA and its receptors are responsible for long-lasting mechanical allodynia and thermal hyperalgesia as well as demyelination and upregulation of pain-related protein expression patterns in a variety of animal models. Evidence supports the hypothesis that antagonism of the LPA signaling pathways would benefit the treatment of neuropathic pain. We have developed the first monoclonal antibody targeting LPA (anti-LPA Ab). In preliminary studies, Lpath's anti-LPA mAb has demonstrated efficacy in the Seltzer model of neuropathic pain where the literature has demonstrated a role for LPA in the disorder, we next turned to diabetic pain models where LPA has not heretofore been implicated. Diabetic neuropathy in type 1 and 2 diabetes in both humans and animal models is characterized by pathophysiological changes in all components (sensory, motor and autonomic) of the peripheral nervous system. In addition to the changes of primary afferent nerves, central sensitization is believed to be an important mechanism underlying persistent pain, including neuropathic and inflammatory pain. The efficacy of our anti-LPA mAbs in the Seltzer model, in part, validates LPA as a target for nerve-injury induced neuropathic pain. We have shown that blocking the activity of LPA with anti-LPA antibodies is sufficient to both prevent the onset of neuropathic pain and alleviate established pain in the rat STZ-induced model of diabetic neuropathy. We next wanted to determine the optimal dose level for the mouse anti-lysophosphatidic acid (LPA) antibody (B3) to reach long term reversal of allodynia. This study was performed in the laboratory of Dr. Nigel Calcutt at UCSD using the well-established rat model of STZ-induced type 1 (insulin deficient)

diabetes. The STZ model uses tactile allodynia as a behavioral index of diabetes-induced neuropathic pain.

3 MATERIALS AND METHODS

3.1 Test Article and Control

Murine anti-LPA antibody (504B3/B3) Lot# HR111266

Murine IgG2b antibody (control IgG/LT1015) Lot# HR91184

3.2 STZ Model Development

Female Sprague-Dawley rats (Harlan Industries, San Diego CA) weighing 225-250 grams each were maintained at room temperature, between 65 to 82°F with relative humidity between 30 to 70%. The room was illuminated with fluorescent lighting on a daily 12 hour light/dark cycle. All animals were maintained 2/cage with free access to dry food and municipal water. Insulin deficient diabetes was induced following an overnight fast by a single IP injection of STZ (55 mg/kg) dissolved in 0.9% sterile saline. Hyperglycemia was confirmed 4 days later and also prior to behavioral testing in a sample of blood obtained by tail prick using a strip operated reflectance meter. All animals were observed daily and weighed regularly during the study period.

3.3 Sample Collection and Analysis

The blood samples were collected in tubes containing potassium EDTA (K₂EDTA) predose (prior to dosing), and 24 hours after last dose each week for the duration of the study. Blood samples were centrifuged at approximately 5000 rpm for approximately 10 minutes at 4°C, and plasma was harvested. Plasma samples were immediately frozen in dry ice and stored at -70 ± 10°C until assayed.

B3 and 1015 antibody plasma concentrations were quantified using a direct-binding immunoassay (ELISA) method. Goat-anti mouse IgG (gamma-chain specific) UNLB antibody (Southern Biotech 1030-01, 1 mg/mL) was diluted 1:1000 in carbonate buffer (100 mM NaHCO₃, 33.6 mM Na₂CO₃, pH 9.5). 384 well plates were coated with 15 µL/well of this coating solution and incubated at 4°C overnight. The plates were then washed 4 times with PBS and blocked with 50 µL/well blocking buffer (1%BSA, 0.1% Tween-20 in PBS) for 1 hour at room temperature. Samples and standard were prepared on non-binding plates with enough volume to run in duplicate. The standard was prepared by diluting mouse anti-LPA antibody in blocking buffer (1%BSA, 0.1% Tween-20 in PBS) to the following dilutions: 900 ng/ml, 300 ng/ml, 100 ng/ml, 33.33 ng/ml, 11.11 ng/ml, 3.70 ng/ml, 1.23 ng/ml, 0.41 ng/ml, 0.14 ng/mL, 0.05 ng/mL, 0.02 ng/mL, and 0 ng/mL. The samples were diluted in blocking buffer 1:10, 1:30, 1:90, 1:270, 1:810, 1:2430, 1:7290, 1:21870, 1:65610, 1:196380 and 1:590490 and 15 µL of the standard/samples preparation

were loaded to each well and incubated at room temperature for 1 hour. Plates were washed 4 times with PBS and 15 μ L of 1: 10,000 blocking buffer diluted goat anti-mouse IgG HRP (Southern Biotech 1030-05) was added to each well and incubated for 1 hour at room temperature. Then, the plates were washed 4 times with PBS and developed using 15 μ L/well TMB substrate at 4°C. After 5-20 minutes, the reaction was stopped with 1M H₂SO₄, 15 μ L/well. The OD was measured at 450 nm. Data were analyzed using Graphpad Prism software. The standard ODs were graphed using a four parameter equation and this was used to calculate the mouse IgG in the samples. Excel software was used to calculate the μ g/ml of mouse IgG in each sample by correcting the values for the dilution factor.

The presence of anti-ADA antibodies were detected by a direct-binding immunoassay (ELISA) method similar to the one described above. B3 and 1015 antibodies were diluted to 1 μ g/mL in carbonate buffer (100mM NaHCO₃, 33.6 mM Na₂CO₃, pH 9.5). 384 well plates were coated with 15 μ L/well of coating antibody solution and incubated at 4°C overnight. The next morning, plates were shook out and 25 μ L/well blocking buffer (1%BSA, 0.1% Tween-20 in PBS) was added and incubated for 1 hour at room temperature. Plates were washed 4 times with PBS, and 15 μ L of diluted test samples per well was added in duplicate and incubate for 1 hour at room temperature. Rat plasma samples were diluted with blocking buffer to 1:10, 1:30, 1:90, 1:270, 1:810, 1:2430, 1:7290, 1:21870, 1:65610, 1:196380 and 1:590490. Then, plates were washed 4 times with PBS and 15 μ L of 1:5,000 blocking buffer diluted goat anti-rat IgG HRP (Jackson 112-035-062) was added to each well and incubate for 1 hour at room temperature. Finally, the plates were washed 4 times with PBS and developed with 15 μ L/well TMB substrate at 4°C. After 5 minutes, the reaction was stopped with 1M H₂SO₄, 15 μ L/well and the OD was measured at 450 nm. Data were analyzed using Graphpad Prism software.

3.4 Tactile Response Threshold

Rats were transferred to a testing cage with a wire mesh bottom and allowed to acclimate. Von Frey filaments (Stoelting, Wood Dale IL) were used to determine the 50% mechanical threshold for foot withdrawal. A series of filaments, starting with one possessing a buckling weight of 2.0g, were applied in sequence to the plantar surface of the right hindpaw with a pressure that causes the filament to buckle. Lifting of the paw was recorded as a positive response and the next lightest filament chosen for the next measurement. Absence of a response after 5 seconds prompted use of the next filament of increasing weight. This paradigm was continued until four measurements were made after an initial change in the behavior or until five consecutive negative (given the score of 15g) or four positive (score of 0.25g) scores occurred. The resulting sequence of positive and negative scores was used to interpolate the 50% response threshold. Only rats with a 50% tactile response threshold below 6g are considered allodynic and brought forward for drug testing.

3.5 Data Analyses

All data were generated and collected at the laboratory of Dr. Nigel Calcutt at UCSD. Descriptive statistics was used to calculate median, mean, standard deviation (SD) and standard error of the mean (SEM) using Microsoft Excel. An excel spread sheet containing all data and analysis was sent via e-mail to Lpath. GraphPad Prism software (version 6.0, Graph Software inc, La Jolla, CA) was used to graph and analyze (ANOVA) the data.

3.6 Study Design

To understand if there is enough mAb at the site of action and/or if there is an improvement in allodynia over time with exposure to B3, a repeat dosing study was designed. Five groups of diabetic rats were established by treatment with STZ and once allodynia was confirmed animals were treated s.c, starting on day 0 with either anti-LPA mAb at 10 or 30 mg/kg or IgG control at 10 or 30 mg/kg two times a week. Control diabetic animals were not treated. Antibody treatment was stopped on day 24 post treatment and allodynia was follow for additional 15 days to day 39 when the study was ended. Mechanical allodynia was measured as described above on days 0, 3, 6, 9, 14, 17, 21, 24, 28, 31, 35 and 39. All measurements were completed 3 days after last antibody treatment.

Table 1 Study Design

Group 1 : Control				
	Antibody/Condition	Vehicle		
	Lot Number	N/A		
	Dose level	same volumen as mAb		
	Dose Route	s.c		
	Frequency	2X Week		
	Total Doses	10		
Group 2 : STZ				
	Antibody/Condition	Vehicle		
	Lot Number	N/A		
	Dose level	same volumen as mAb		
	Dose Route	s.c		
	Frequency	2X Week		
	Total Doses	10		
Group 3 : STZ+anti-LPA 30mg/kg				
	Antibody/Condition	504B3		
	Lot Number	HR911266		
	Dose level	30 mg/kg		
	Dose Route	s.c		
	Frequency	once		
	Total Doses	10		
Group 4 : STZ+IgG 30mg/kg				
	Antibody/Condition	LT1015		
	Lot Number	HR91184		
	Dose level	30 mg/kg		
	Dose Route	s.c		
	Frequency	2X Week		
	Total Doses	10		
Group 5 :STZ+anti-LPA 10mg/kg				
	Antibody/Condition	504B3		
	Lot Number	HR911266		
	Dose level	10mg/kg		
	Dose Route	s.c		
	Frequency	2X Week		
	Total Doses	10		
Group 6 : STZ+IgG 10mg/kg				
	Antibody/Condition	LT1015		
	Lot Number	HR91184		
	Dose level	10mg/kg		
	Dose Route	s.c		
	Frequency	2X Week		
	Total Doses	10		

4 RESULTS

4.1 Body Weight

Diabetic rats had a lower weight than control rats by the end of the study. The mean ($\pm$ SEM) body weight for diabetic rats at conclusion of the study (284 ± 10 g) was higher than the corresponding mean ($\pm$ SEM) body weight of the control group (216 ± 16 g) (**p < 0.01 by one-way ANOVA with Bonferroni post-hoc analysis). There was no significant difference in weight between, vehicle and B3 antibody treated rats (Table 2).

Table 2 Body Weight and Blood Glucose

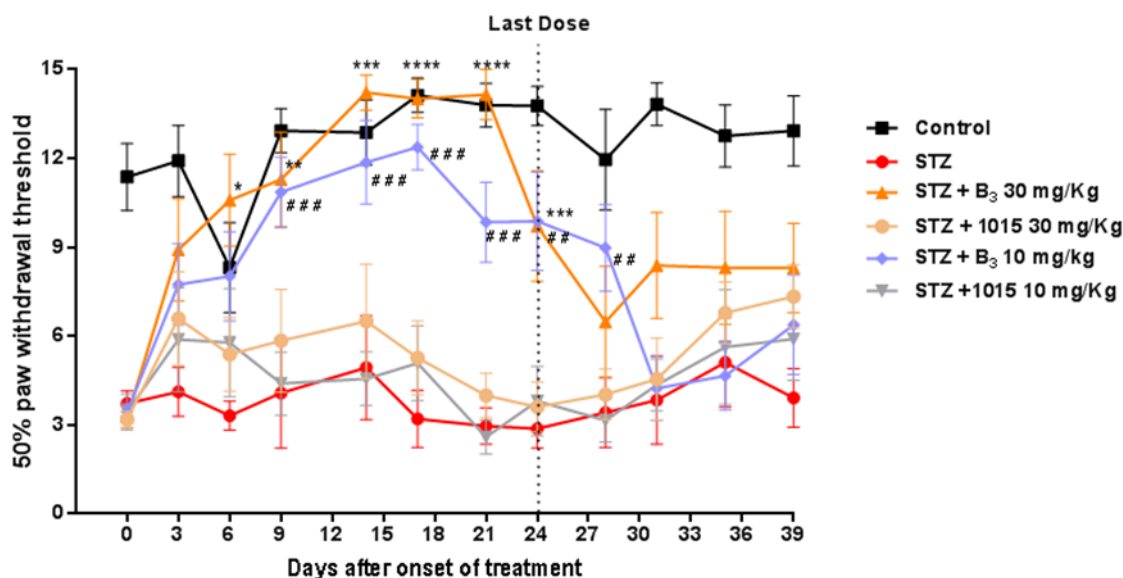
GROUP	Body weight		Blood glucose			GROUP	Body weight		Blood glucose		
Control	Onset	Terminal	Onset	Terminal			Onset	Terminal	Onset	Terminal	
1.0	235	276				STZ+IgG 30mg/kg					
1.1	237	292				15.0	233	161	538	551	
1.2	230	290				15.1	230	157	568	599	
2.0	248	303				16.0	216	170	500	546	
2.1	232	281				16.1	228	213	534	599	
2.2	241	273				21.1					
3.0	235	294				22.0					
3.1	241	271				7.0	222	189	500	465	
4.0	225	285				7.1	220	242	543	565	
4.1	236	278				19.0	233	207	584	572	
MEAN	236	284				19.1	250	204	454	554	
SD	6	10				MEAN	229	193	528	556	
SEM	2	3				SD	11	29	42	42	
STZ						SEM	4	10	15	15	
5.0	228	256	429	594		STZ+anti-LPA 10mg/kg					
5.1	225	DIED	398	DIED		20.0	233	268	483	599	
17.0				501		20.1	232	221	570	599	
17.1	223	182	574	599		21.0	237	241	550	529	
18.0						6.0	235	200	511	585	
18.1	230	212	599	478		6.1	237	280	499	542	
8.0	221	204	442	577		22.1	232	194	542	544	
8.1	222	263	455	472		23.0	225	177	449	599	
9.0						23.1	245	255	561	599	
9.1	233	179	485	599		24.0	247	257	474	404	
MEAN	226	216	483	546		24.1	220	186	490	474	
SD	5	36	76	59		MEAN	234	228	513	547	
SEM	2	15	29	22		SD	8	37	41	65	
STZ+anti-LPA 30mg/kg						SEM	3	12	13	21	
10.0	222	185	504	507		STZ+IgG 10mg/kg					
10.1	226	275	461	449		25.0	245	199	477	599	
11.0	228	DIED	529	DIED		25.1					
11.1	234	219	458	594		26.0	249	295	596	536	
12.0	246	212	522	582		26.1	236	198	497	599	
12.1	231	231	504	470		27.0	227	242	489	566	
13.0	234	274	573	472		27.1	223	202	515	579	
13.1	234	220	589	599		28.0	235	209	485	582	
29.0						28.1	232	183	493	449	
29.1	247	183	564	599		14.0	238	132	463	482	
MEAN	234	225	523	534		14.2	222	222	501	516	
SD	8	35	47	66		MEAN	234	209	502	545	
SEM	3	12	16	23		SD	9	44	38	54	
						SEM	3	15	13	18	

4.2 Blood Glucose

Blood glucose concentration increased approximately 5-fold in diabetic rats compared with reference values for female Sprague-Dawley rats ([Table 2](#)) (**** $p < 0.0001$ by two-way repeated measures ANOVA with Bonferroni *post-hoc* analysis). There was no significant difference in blood glucose between STZ animals treated with vehicle, IgG control or anti-LPA antibody ([Table 2](#)).

4.3 Mechanical Allodynia

[Figure 1](#) shows that the anti-LPA antibody (at 10 mg/kg, blue diamond or 30 mg/kg, orange triangle) increased the paw withdrawal threshold in diabetic rats when compared with IgG control (at 10 mg/Kg silver triangle, or 30 mg/kg tan circle) or vehicle treated animals (red circle) to a level similar to that observed in non-diabetic rats (black squares) ($^{##}p < 0.01$, $^{###}p < 0.001$ v. LT1015 at 10 mg/Kg and $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$ v. LT1015 at 30 mg/Kg by two-way repeated measures ANOVA with Bonferroni *post-hoc* analysis). This marked reduction in mechanical allodynia is observed 3 days after drug delivery (when allodynia was measured) indicating a sustained and specific effect of the antibody. In order to determine the washout time of the antibody in circulation and the possible effect on efficacy measured as mechanical allodynia, on day 28 of the experiment all drug treatments were stopped while mechanical allodynia continued to be measured weekly to the end of the study (day 39). As expected, [Figure 1](#) shows a progressive decay in PWT that corresponds to a decrease in circulating levels of anti-LPA antibody ([Table 3](#)). The reduction in effect may also have been the result of rat anti-mouse antibody generated following repeat dosing of B3 ([Appendix 1](#)).

Figure 1. Effect of Lpathomab on Diabetic Induced Allodynia.

Tactile allodynia was measured in Control (black square) and 5 weeks diabetic (allodynic) rats. Allodynic animals were treated s.c, starting on day 0 with either B3 (at 10 mg/kg, blue diamond or 30 mg/kg, orange triangle) or 1015 control (at 10 mg/kg, silver triangle, or 30 mg/kg tan circle) two times a week. Control diabetic animals (red circle) were not treated. Antibody treatment was stopped on day 24 post treatment and allodynia was followed to the end of the study (day 39). Mechanical hypersensitivity (tactile allodynia) was assessed by Von Frey filaments according to the "up-down method" and the data were processed to calculate 50% probability to get a withdrawal response. Mechanical allodynia was assayed 3 days after dose at all time. Each time point represents mean $\pm$ SEM, n=7-10 rats/group. ##p < 0.01, ###p < 0.001 v. LT1015 at 10 mg/Kg and **p < 0.01, ***p < 0.001, ****p < 0.0001 v. LT1015 at 30 mg/Kg by two-way repeated measures ANOVA with Bonferroni post-hoc analysis.

Table 3 Antibody Concentrations

mAb [$\mu\text{g/ml}$] After last dose (before wash out)			
B3 -30 mg/kg			B3 -10 mg/kg
Sample	Mean	SDev	Sample Mean SDev
13.1	234	30	20.1 3 0
13.0	11	0	24.0 23 2
12.1	278	27	20.0 18 0
12.0	99	36	23.1 4 0
11.1	97	4	23.0 102 28
10.1	5	3	22.1 185 18
29.1	288	2	6.1 8 2
10.0	297	16	6.0 70 9
29.0	243	26	21.0 13 1
1015 -30 mg/kg			1015 -10 mg/kg
Sample	Mean	SDev	Sample Mean SDev
21.1	262	48	27.1 96 15
16.1	451	162	27.0 25 1
16.0	464	104	26.1 94 1
15.1	250	19	26.0 97 16
15.0	477	128	14.2 108 0
19.1	368	34	25.1 140 12
19.0	78	6	14.0 63 14
7.1	278	36	25.0 76 8
7.0	378	28	28.1 137 31
22.0	312	51	28.0 145 3

mAb [µg/ml] After last dose

B3 -30 mg/kg

Sample	Mean	SDev
13.1	0.69	0.08
13.0		
12.1	2.49	0.47
12.0	0.11	0.02
11.1	0.08	0.01
10.1		
29.1	0.16	0.03
10.0	4.46	0.22
29.0	0.08	0.00

B3 -10 mg/kg

Sample	Mean	SDev
20.1		
24.0	0.06	0.01
20.0		
23.1		
23.0	0.89	0.09
22.1	1.76	0.38
6.1		
6.0	0.05	0.01
21.0		
24.1	0.05	0.01

1015 -30 mg/kg

Sample	Mean	SDev
21.1	0.20	0.03
16.1	25.45	5.40
16.0	20.69	4.14
15.1	0.29	0.02
15.0	14.58	2.50
19.1	4.30	0.34
19.0	0.18	0.02
7.1	0.06	0.02
7.0	5.64	0.25
22.0	8.35	1.63

1015 -10 mg/kg

Sample	Mean	SDev
27.1	0.25	0.00
27.0		
26.1		
26.0	0.96	0.08
14.2	4.82	0.64
25.1	4.74	0.28
14.0	0.20	0.03
25.0	0.40	0.04
28.1	6.20	0.54
28.1	1.63	0.48

5 CONCLUSIONS

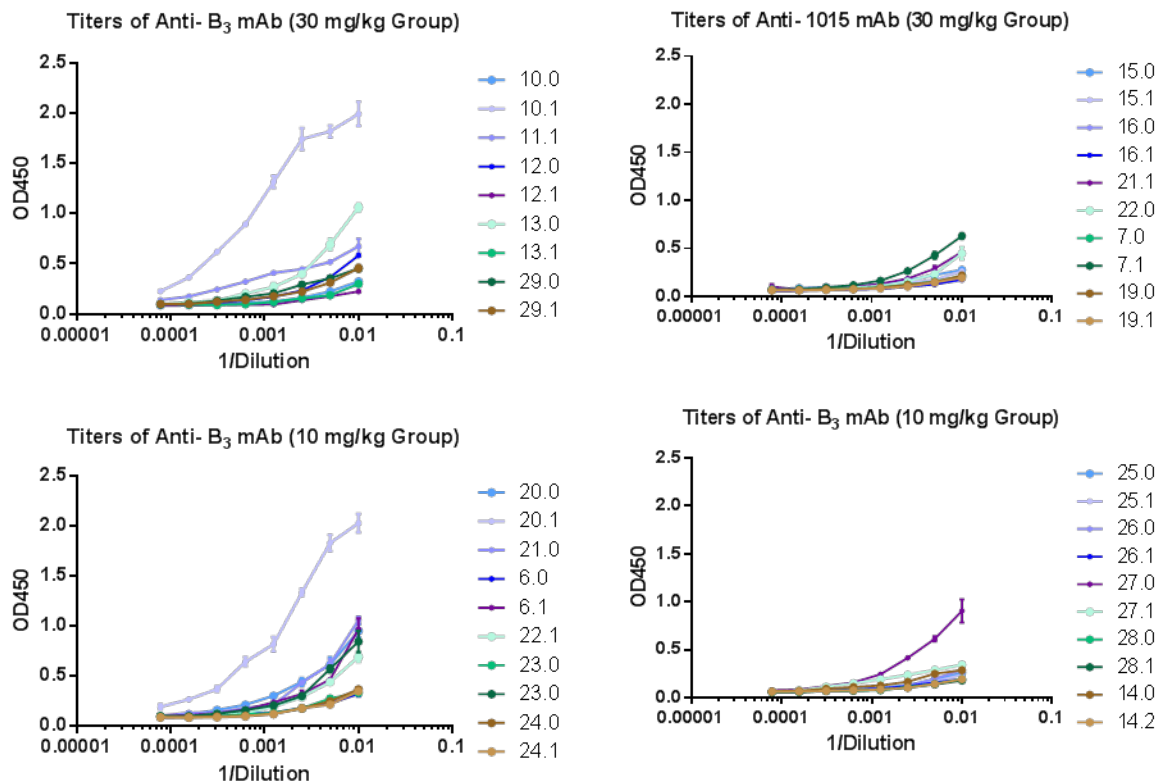
Repeat doses of B3 at 10 and 30 mg/kg increased the paw withdrawal threshold in diabetic rats to a level similar to that observed in non-diabetic rats, indicating a reversal of established allodynia in rats with diabetes-induced neuropathic pain.

Once B3 therapy was stopped, the anti-allodynic effect of the antibody washed out, and over time the diabetic rats returned to be allodynic.

6 DATA ARCHIVE

LPath Notebook # 137 pp. 23-25, 48-56; #248 pp. 47-51

Appendix 2 Detection of Anti-B3 and Anti-1015 in Plasma



Anti-B₃ and Anti-LT1015 antibody titers. B₃ and LT1015 antibodies were diluted to 1 µg/mL to coat plates and the ELISA was carried out as described in the methods. Rat plasma samples from B₃ treated group were diluted with blocking buffer to 1:10, 1:30, 1:90, 1:270, 1:810, 1:2430, 1:7290, 1:21870, 1:65610, 1:196380 and 1:590490. Data was analyzed using Graphpad Prism software.